

MANAS AGRAWAL

Boston, MA | (857) 265-1533 | aggrawal.m@northeastern.edu | [Portfolio](#) | [Linkedin](#) | [Github](#) | [Medium](#) | [Leetcode](#)

EDUCATION

Northeastern University

Masters of Science – Computer Science | GPA: 3.86

Boston, MA

Sep 2024 – Dec 2026

- Graduate Teaching Assistant for Fundamentals of Software Engineering (CS 4530)
- Researching Multi-Agent LLM architectures, for e2e SDLC automation, across accuracy, cost efficiency

WORK EXPERIENCE

Hydrow

Software Engineer Intern

Boston, MA

Jan 2026 – Aug 2026

- Owned several product features end-to-end — schema, APIs, tests, and rollout — including the strength-fitness app's progress graphs, keeping every user-facing endpoint under 25 ms at production scale.
- Drove large latency wins by pushing data-heavy work into SQL and caching hot paths — one graph computation went from an event-loop-blocking loop to 0.7 ms, and real-time checks from ~2,000 ms to near-zero with materialized views and Redis.
- Built the strength recommendation engine — a multi-pass SQL pipeline that only suggests workouts a user has the equipment and difficulty for, relaxing constraints step by step so it never runs out of options.
- Shipped the real-time gamification system — 5+ badge checkers plus a schema API that generates admin forms for all 21 badge types straight from their validation schemas, with no frontend changes.
- Designed a pluggable A/B experiment framework — a master/leaf tile mechanism that serves different user groups different content, targeted by churn score and behavioral segment.
- Delivered a full image-banner CRUD backend on S3, ordering the storage and database writes with rollback so a failed write never leaves an orphaned file.
- Restored data correctness across ~123K historical records with a bounded, dry-run-verified backfill migration, and kept large-table reads scale-safe with cursor-based pagination.
- Built an AI-assisted development workflow on Claude Code that automates the coding, testing, and self-review stages of the SDLC while keeping me in the loop for planning and final review, backed by a persistent knowledge base of accumulated review standards.

canAssist × Northeastern University

Product Engineer & Scrum Master, Research Capstone

Boston, MA

May 2026 – Aug 2026

- Led a 5-person team — drove product features, client requirements, sprint ceremonies, and demos while building a NestJS/PostgreSQL/Redis backend instrumented with my own backend boilerplate and published telemetry package

Studio Graphene

Forward Deployed Engineer

Gurgaon, India

Nov 2020 – Jul 2024

- Built a serverless engineering-analytics platform on AWS Lambda to pinpoint bottlenecks (PR wait times, build failures, blocked dependencies), boosting engineering velocity by 23%
- Designed a Node.js/TypeScript REST backend serving 2M+ daily requests, with PostgreSQL partitioning/indexing to handle a 500GB+ catalog and real-time cross-region inventory sync
- Built event-driven microservices with AWS SQS for decoupled ingestion and processing, reducing response latency by ~40%

OPEN SOURCE PROJECTS

[Telemetry NPM Package](#) | [Nodejs](#), [TypeScript](#), [AWS X-Ray](#), [npm](#), [OpenTelemetry](#), [cloudwatch](#), [Jaeger](#), [Prometheus](#)

- A vendor-agnostic OpenTelemetry package with pluggable presets for AWS, Honeycomb, Jaeger, Tempo, or any OTLP-native backend. Ships traces, metrics, and logs through a single sidecar collector with cost-mindful defaults thereby reducing mean debugging time by ~60%
- It has over 1000 downloads on npm

TECHNICAL SKILLS

Languages:

JavaScript, TypeScript, Java, C++, Python

Frameworks:

Node.js, Express.js, NestJS, Django, Laravel, React

Database:

SQL, Postgres, MySQL, NoSQL, MongoDB, Elasticsearch, Amazon DynamoDB, Firebase

Cloud & DevOps:

Cloud Computing, AWS (Lambda, S3, SQS, DLQ, X-Ray), Docker, CI/CD, cloud infrastructure

Others:

Git, Sentry, OpenTelemetry, linux, REST APIs, web architecture, frontend, agile, SDLC